

Thermodynamics

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Q1. Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is ____.

NTA B.Tech · 2 Apr 2026, Shift 1 (Q33)

- (1) 2:3:5
- (2) 5:3:2
- (3) 2:5:7
- (4) 7:5:2

✓ **Answer: (4) 7:5:2**

Solution:

For a diatomic gas, $C_v = 5R/2$ and $C_p = 7R/2$.

$$\Delta Q = nC_p \Delta T = 7R\Delta T/2$$

$$\Delta U = nC_v \Delta T = 5R\Delta T/2$$

$$\Delta W = \Delta Q - \Delta U = R\Delta T$$

$$\text{Ratio} = 7/2 : 5/2 : 1 = 7 : 5 : 2.$$

Q2. A vessel contains 0.15 m^3 of a gas at pressure 8 bar and temperature 140°C with $C_p = 3R$ and $C_v = 2R$. It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is ____ kJ.

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✓ **Answer: 120**

Solution:

$$\gamma = C_p/C_v = 3R/2R = 1.5.$$

$$\text{Using } PV^\gamma = \text{constant: } V_2 = V_1(P_1/P_2)^{1/\gamma} = 0.15 \times 8^{2/3} = 0.15 \times 4 = 0.6 \text{ m}^3.$$

$$\begin{aligned} \text{Work done } W &= (P_1 V_1 - P_2 V_2)/(\gamma - 1) = (8 \times 10^5 \times 0.15 - 1 \times 10^5 \times 0.6)/0.5 \\ &= (120000 - 60000)/0.5 = 120000 \text{ J} = 120 \text{ kJ}. \end{aligned}$$

Source: NTA B.Tech — 2 April 2026, Shift 1 (official NTA release, physics section) · formulaverse.in